Practical 4: Uploading Data to CRD

Day 2 — Duration: 1.5 Hours

SDI Training for ICT Professionals — Climate Risk Database

What you will accomplish: By the end of this practical, your CRD will contain at least two geographic datasets with proper metadata. You will also connect QGIS to your CRD and view the uploaded layers on your desktop GIS.

Before you start: Your CRD must be running and you must be able to log in as admin at `http://localhost`. You also need the sample data files provided by the instructor.

Step 1: Get the Sample Data Files

Step 1.1: The instructor will provide sample data files via one of these methods:

- **USB drive:** Copy the files to your `Desktop` or `Downloads` folder
- **Shared folder:** Download from the URL the instructor provides
- **Direct download from CRD:** Download from the live CRD platform

Step 1.2: You should have these files on your computer:

- `mwanza_businesses.geojson` — Business locations in Mwanza (vector/point data)
- `tanga_wards.zip` — Ward boundaries as a Shapefile (vector/polygon data)
- `sample_elevation.tif` — A GeoTIFF elevation raster (raster data)

Step 1.3: Place all files in an easy-to-find folder, for example your `Desktop`.

Data format quick reference:

- **GeoJSON:** A single `.geojson` file containing vector data (points, lines, or polygons). Easy to use — just one file.
- **Shapefile:** A set of files (`.shp`, `.shx`, `.dbf`, `.prj`) that together describe vector data. Must be uploaded as a `.zip` archive.
- **GeoTIFF:** A `.tif` file containing raster data (like satellite images or elevation grids). Each pixel has a value.

Step 2: Upload a GeoJSON File via the Web Interface

Step 2.1: Open your browser and go to your CRD:

```
http://localhost
```

Step 2.2: Make sure you are logged in as admin. If not, click **"Sign in"** and enter your credentials.

Step 2.3: In the top navigation menu, click **"Data"** (or **"Upload"** if available).

Step 2.4: Click the **"Upload Layer"** or **"Upload Dataset"** button.

You should see: An upload page with a large area where you can drag and drop files, or a **"Choose Files"** button.

Step 2.5: Either drag the file `mwanza_businesses.geojson` into the upload area, or click **"Choose Files"** and navigate to where you saved the file.

You should see: The filename appears in the upload area, ready to be uploaded.

Step 2.6: Click the **"Upload"** button.

You should see: A progress bar. After a few seconds, the upload completes and you are taken to the layer's detail page (or a success message appears).

If the upload fails:

- Check that the file is valid GeoJSON (not corrupted)
- Check that the file is not too large (under 100 MB for most configurations)
- Check the Docker logs: `docker compose logs django -tail 30`

Step 3: Set Metadata

Metadata describes your data — what it is, who created it, when, and where. Good metadata makes data discoverable and useful.

Step 3.1: After uploading, you should be on the layer's detail page. Click **"Edit Metadata"** (or the pencil icon).

You should see: A form with many fields for describing your dataset.

Step 3.2: Fill in the following fields:

- **Title:** `Mwanza Business Locations`
- **Abstract:** `Point locations of businesses in Mwanza City, Tanzania. Collected as part of the community resilience mapping project. Includes business name, type, and GPS coordinates.`
- **Keywords:** Type and add each keyword separately: `mwanza`, `businesses`, `economic`, `resilience`
- **Category:** Select **"Economy"** or the most appropriate category from the dropdown
- **Region:** Select **"Africa"** → **"Tanzania"** if available
- **Language:** Select **"English"**

Step 3.3: Click **"Save"** or **"Update"**.

You should see: The detail page now shows your title, abstract, and keywords. The layer is now properly documented.

Why metadata matters: Without metadata, a dataset is just a file. With metadata, people can find it by searching, understand what it contains, know when it was created, and decide if it is useful for their work. Always fill in at least the title, abstract, and keywords.

Step 4: Set Permissions

Step 4.1: On the layer's detail page, look for a **"Share"** or **"Permissions"** button and click it.

You should see: A permissions panel or dialog showing who can view, download, and edit this layer.

Step 4.2: Set the following permissions:

- **Anyone (public):** Can **view** and **download**
- **Registered users:** Can **view** and **download**
- **Only you (admin):** Can **edit** and **manage**

Step 4.3: Click **"Save"** or **"Apply"**.

Step 4.4: To verify: open a private/incognito browser window, go to `http://localhost`, and check that the "Mwanza Business Locations" layer appears in the data list without logging in.

You should see: The layer is visible to anyone, even without an account.

Step 5: View the Data on the Map

Step 5.1: On the layer's detail page, click **"Map View"** or **"View on Map"**.

You should see: An interactive map showing your business location points overlaid on a base map. You can zoom in and out, and click on points to see their attributes (name, type, etc.).

Step 5.2: Try clicking on individual points on the map.

You should see: A popup or sidebar showing the attributes of that point — such as the business name, type, and address.

Step 5.3: Try the **"Identify"** or **"Info"** tool if available. Click on a point to see all its attributes.

Step 6: Upload a Shapefile

Shapefiles are the most common format in GIS. Unlike GeoJSON (one file), a Shapefile consists of multiple files that must be uploaded together as a ZIP archive.

Step 6.1: First, understand what is inside a Shapefile:

- `.shp` — The shapes (geometry) — points, lines, or polygons
- `.shx` — An index file that helps software read the shapes quickly
- `.dbf` — The attribute table (data about each shape)
- `.prj` — The coordinate system definition (where on Earth the shapes are)

All four files are **required**. Without any one of them, the upload will fail.

Step 6.2: The instructor has provided these files already packaged as `tanga_wards.zip`. If you have the individual files instead, you need to create a ZIP:

Windows: Select all four files, right-click → **"Compress to ZIP file"**

Mac: Select all four files, right-click → **"Compress 4 Items"**

Ubuntu:

```
zip tanga_wards.zip tanga_wards.shp tanga_wards.shx tanga_wards.dbf tanga_wards.prj
```

What this does: Combines all four Shapefile component files into a single ZIP archive that CRD can process.

You should see: A new file called `tanga_wards.zip` appears.

Step 6.3: Go back to your CRD: `http://localhost`

Step 6.4: Click **"Data"** → **"Upload Layer"** (or **"Upload Dataset"**).

Step 6.5: Drag the `tanga_wards.zip` file into the upload area, or click **"Choose Files"** and select it.

Step 6.6: Click **"Upload"**.

You should see: A progress bar followed by a success message. CRD automatically extracts the ZIP and processes the Shapefile.

Step 6.7: Fill in the metadata:

- **Title:** `Tanga Ward Boundaries`
- **Abstract:** `Administrative ward boundaries for Tanga City, Tanzania.`
- **Keywords:** `tanga`, `wards`, `administrative`, `boundaries`
- **Category:** `Boundaries` or `Administrative`

Step 6.8: Set permissions to public view (same as Step 4).

Step 6.9: Click **"Map View"** to see the ward polygons on the map.

You should see: Coloured polygon shapes showing the ward boundaries of Tanga, overlaid on the base map.

Common Shapefile upload errors:

- **"Missing .prj file":** The coordinate system file is missing. Ask the data provider for it, or create one in QGIS.
- **"Encoding error":** The .dbf file contains special characters. Try adding a `.cpg` file containing the text `UTF-8` to the ZIP.
- **"File too large":** The ZIP exceeds the upload limit. Ask the instructor to adjust the limit or simplify the data.

Step 7: Connect QGIS to Your CRD

Now that you have data on your CRD, let us view it in the professional QGIS desktop application using a WMS (Web Map Service) connection.

Step 7.1: Open **QGIS** on your computer.

Step 7.2: In the top menu, click **Layer** → **Add Layer** → **Add WMS/WMTS Layer...**

You should see: A dialog window titled "Data Source Manager — WMS/WMTS".

Step 7.3: Click the **"New"** button to create a new connection.

Step 7.4: Fill in the connection details:

- **Name:** `My Local CRD`
- **URL:**

```
http://localhost/geoserver/ows
```

What this does: Tells QGIS where to find the map server (GeoServer) that serves your CRD layers as images.

Step 7.5: Click **"OK"** to save the connection.

Step 7.6: Make sure **"My Local CRD"** is selected in the dropdown, then click **"Connect"**.

You should see: After a few seconds, a tree of available layers appears. You should see your uploaded layers listed (e.g., "Mwanza Business Locations", "Tanga Ward Boundaries").

Step 7.7: Click on a layer to select it (e.g., "Mwanza Business Locations").

Step 7.8: Click **"Add"** at the bottom of the dialog.

You should see: The layer appears on the QGIS map canvas. Points or polygons are drawn on the map.

Step 7.9: Add the second layer by selecting it and clicking **"Add"** again.

Step 7.10: Click **"Close"** to close the dialog.

You should see: Both layers in the QGIS Layers panel on the left. The map canvas shows your data. You can zoom in, zoom out, and click to identify features.

WMS vs WFS — what is the difference?

- **WMS (Web Map Service):** Sends map *images* to QGIS. Good for viewing, but you cannot edit the data or run analysis.
- **WFS (Web Feature Service):** Sends the actual *data* (geometries and attributes) to QGIS. You can query, filter, and analyse the data.

We used WMS here for simplicity. In Practical 7, you will use WFS for more advanced work.

Checkpoint

Before proceeding to Practical 5, confirm:

☐ **GeoJSON uploaded:** "Mwanza Business Locations" is visible in your CRD

☐ **Shapefile uploaded:** "Tanga Ward Boundaries" is visible in your CRD

☐ **Metadata complete:** Both layers have titles, abstracts, and keywords

☐ **Permissions set:** Both layers are publicly viewable

☐ **Map view works:** You can see both layers on the CRD map viewer

☐ **QGIS connected:** At least one layer loads in QGIS via WMS

Great progress! You now know how to upload geographic data to CRD and connect desktop GIS software to it. Tomorrow you will learn to collect field data with KoboToolbox and integrate it with CRD.

End of Practical 4 — Proceed to Practical 5: Installing KoboToolbox & Designing a Survey